

Homework4 CS5800

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1 Exercise

(a)

If we had groups of size 3 instead, we get $\frac{n}{3}$ groups. So finding the median of each group would require sorting $\frac{n}{3}$ groups of size 3. To find representatives it would take $\frac{n}{3}O(1) = O(n)$ and to find the median of representatives it would take $T(\frac{n}{3})$. In this case we now there will be only 1 value lower than x (pivot) and 1 higher, in each group. The total number of elements higher than x and of the ones lower than x will be $\geq 2\frac{n}{6}$ ($\frac{n}{6}$ blocks). After partitioning, we recurse on at most $\frac{4n}{6}$ elements. So in total we would get:

$$T(n) \leq \Theta(n) + T(\frac{n}{3}) + T(\frac{4n}{6})$$

(b)

For size $c \geq 5$ and odd, we would get:

$$T(n) \leq \Theta(n) + T(\frac{n}{c}) + T(1 - \lceil \frac{c}{2} \rceil \frac{n}{c})$$

($\frac{n}{c}$ blocks, and number of elements higher than the pivot x or lower than x being $\geq \lceil \frac{c}{2} \rceil$).

(c)

case 1 (groups of size $\sqrt{n}$):

The total number of groups is $\sqrt{n}$ and each contains $\sqrt{n}$ elements. The total time to find the median would be $O(\sqrt{n}\sqrt{n}) = O(n)$. Now when recursively applying the selection algorithm on the medians of the $\sqrt{n}$ groups, each recursive step reduces the problem size by a factor of $\sqrt{n}$, so the number of recursive steps would be $\log_{\sqrt{n}} n = 2 \log_2 n$. So in general, $T(n) = O(n) * O(\log n) = O(n \log n)$.

case 2 (5 groups of size $\frac{n}{5}$):

Since there are 5 groups, the total time for finding the medians is $O(5\frac{n}{5}) = O(n)$. We would apply selection algorithm on 5 medians which can be done in constant time since 5 is a small fixed number. $T(n) = T(\frac{n}{5}) + O(n) = O(n)$.